DEMONSTRATION OF PROPOSED FEATURES

Date	25 September 2026
Team ID	SWTID-2026-4307
Project Name	AI Weather Wise API
Maximum Marks	1 Mark

Demonstration of Proposed Features:

> *This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.*

S.No	Feature Name	Description	Status	Demonstrated	Remarks
1	**Real-Time Weather Data Retrieval**	Fetches real-time weather metrics (temperature, humidity, pressure, wind speed) via API endpoints.	Implemented	Yes	Successfully integrated and tested with live data inputs.
2	**AI-Driven Weather Forecasting**	Uses Machine Learning models to predict short-term and long-term weather trends and anomalies.	Implemented	Yes	Predictions generated accurately with high performance.
3	**Location-Based Query Support**	Enables users to query weather insights using geolocations, zip codes, or city names.	Implemented	Yes	Dynamic geolocation lookup working as intended.
4	**Severe Weather Alert Notifications**	Automatically detects extreme weather conditions and triggers automated alert responses.	Implemented	Yes	Threshold-based notifications verified during testing.
5	**Historical Weather Data Insights**	Provides historical weather trends and data logs for analytical and comparative purposes.	Implemented	Yes	Successfully retrieves historical logs from database.
6	**API Authentication & Rate Limiting**	Implements Secure API Key generation, JWT authentication, and rate-limiting middleware.	Implemented	Yes	Fully functional and secured against excessive requests.

Feature Implementation Summary:

Total Features Proposed	**6**
Total Features Implemented	**6**
Total Features Demonstrated	**6**
Overall Implementation Rate (%)	**100%**